

Problems 4

1. Find Z parameters for the circuit shown in Figure 1.

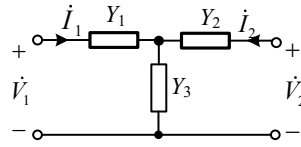


Figure 1

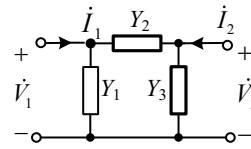
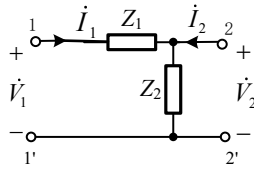
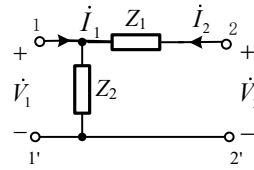


Figure 2

2. Find Y parameters for the circuit shown in Figure 2.
 3. Find T parameters for the circuit shown in Figure 3. What's the relationship between T_a in Figure3(a) and T_b in Figure3(b).



(a)



(b)

Figure 3

4. Find T parameters for the symmetrical T type circuit shown in Figure 4. Using these parameters, determine V_1 and I_1 when $V_2=3V$ and $I_2=-3A$.

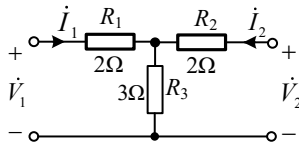


Figure 4

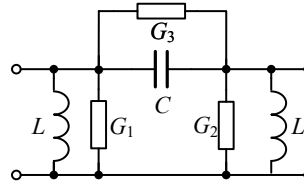


Figure 5

5. Taking advantage of connection in parallel between two-port networks, separate the circuit shown in Figure 5 into a resistor network connected in parallel with a reactor network. Obtain the total Y parameters.
 6. Taking advantage of connection in cascade between two-port networks, separate the circuit shown in Figure 6 into networks connected in cascade. Obtain the total T parameters.

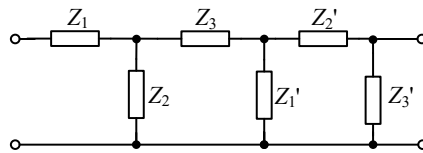


Figure 6

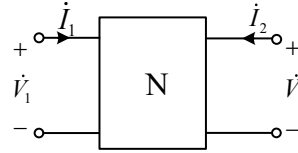


Figure 7

7. The circuit shown in Figure 7 is given $y_{12} = y_{21}$. If

$$z_{01} = \left. \frac{V_1}{I_1} \right|_{V_2=0} = 2\Omega, \quad z_{02} = \left. \frac{V_2}{I_2} \right|_{V_1=0} = \frac{3}{2}\Omega, \quad z_{\infty 1} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = 8\Omega, \quad z_{\infty 2} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 6\Omega,$$

find Y parameter equations.

8. For the circuit shown in Figure 8, (1) find the input current I if $R_L = 1\Omega$; (2) find the transfer voltage ratio $H_V = V_2 / V_1$.

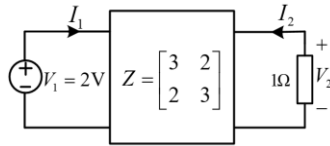


Figure 8

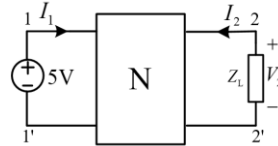


Figure 9

9. For the two-port network shown in Figure 9, when load Z_L at terminals $22'$ is open circuit, we have $I_1 = 1A$, $V_2 = 3V$; when load Z_L at terminals $22'$ is short circuit, we have $I_1 = 25/16A$, $I_2 = -15/16A$. (1) Find T parameter equations; (2) When $Z_L = 4\Omega$, find I_1 , I_2 and V_2 .
10. If the two networks shown in Figure 10 are equivalent, find the relationships needed.

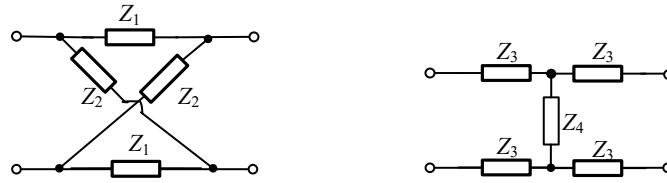


Figure 10

11. Find the equivalent relationships between the two networks shown in Figure 11.

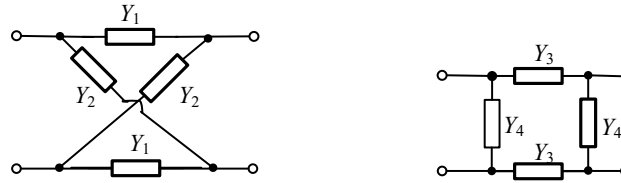


Figure 11

12. Find V_L in the circuit shown in Figure 12 when $R_L = \infty$ and $R_L = 0.25\Omega$, respectively.

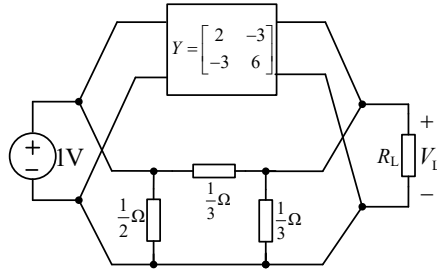


Figure 12

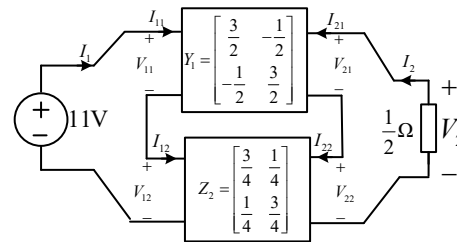


Figure 13

13. Find the current I_1 shown in Figure 13.
14. For the network shown in Figure 14, (1) find the input resistance R_{in} , V_1 and I_1 , at terminals $11'$; (2) obtain the Thevenin equivalence seen from terminals $22'$, and find V_2 and I_2 .

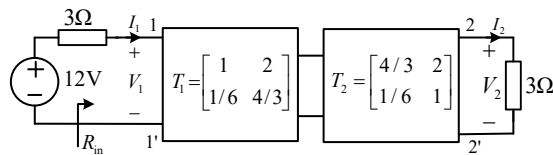


Figure 14